12TH BIOLOGY · CHAPTER 25 · SOLVED EXERCISE

Ecology

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

The numbering of this exercise is kept exactly as printed in the textbook: MCQs 35–44, Short Questions 1–12, Long Questions 31–37 and Inquisitive Questions 18–22.

SECTION 1: MULTIPLE CHOICE QUESTIONS

35. If you study the interaction between a herd of Markhors, the grass they eat, and the snow leopards that hunt them, which level of ecological organization are you focusing on?

(a) Species (b) Population
(c) Community (d) Biosphere

Answer: (c) Community — Different populations of various species interacting in the same area form a community.

36. Nitrogen-fixers are present in the group/s;

(a) Bacteria and fungi (b) Bacteria only
(c) Fungi only (d) Bacteria, fungi and viruses

Answer: (b) Bacteria only — All the nitrogen-fixers named in the chapter (cyanobacteria, *Azotobacter*, *Clostridium*, *Rhizobium*) are bacteria.

37. In an ecosystem, the energy flow is;

(a) Always unidirectional (b) Always bidirectional
(c) Non-directional (d) Cyclic

Answer: (a) Always unidirectional — Unlike the cycling of matter, the flow of energy is unidirectional.

38. Absorption and re-emission of heat radiation by the atmosphere is called;

(a) Global warming (b) Ozone depletion
(c) Acid rain (d) Greenhouse effect

Answer: (d) Greenhouse effect — Greenhouse gases trap the heat radiated from the Earth’s surface.

39. A population’s carrying capacity.

(a) Can be accurately calculated
(b) Generally, remains constant over time
(c) May change as environmental conditions change
(d) Can never be exceeded

Answer: (c) May change as environmental conditions change — It depends on limiting factors (food, water, shelter, predators, diseases), which change with the environment.

studysapiens.com

40. During energy flow in ecosystems, how much of the energy at a trophic level enters the next level?

(a) 100% (b) 50%
(c) 25% (d) 10%

Answer: (d) 10% — Only about 10% is transferred to the next trophic level.

41. Which group of bacteria is involved in nitrification?

(a) Azotobacter (b) Nitrosomonas
(c) Clostridium (d) Rhizobium

Answer: (b) Nitrosomonas — *Nitrosomonas* convert ammonia to nitrite; the other three are nitrogen-fixers.

42. Which step of nitrogen cycle is accomplished by Pseudomonas?

(a) Ammonification (b) Nitrification
(c) Nitrogen fixation (d) De-nitrification

Answer: (d) De-nitrification — *Pseudomonas* and *Thiobacillus* convert nitrates back into nitrogen gas.

43. How do nitrogen-fixing bacteria contribute to the nitrogen cycle?

(a) Return nitrogen (N_2) to the atmosphere
(b) Change ammonium to nitrate
(c) Change N_2 to ammonium
(d) Absorb nitrate from the soil

Answer: (c) Change N_2 to ammonium — Nitrogen fixation converts atmospheric N_2 into ammonia.

44. According to 2025-2026 climate data, the "Enhanced Greenhouse Effect" is considered as climate injustice for Pakistan because;

(a) Pakistan produces the majority of global greenhouse gases.
(b) Greenhouse Effect only occurs in South Asia.
(c) Pakistan emits less than 1% of global GHGs but suffers extreme warming and floods.
(d) Industrialized nations have successfully stopped all carbon emissions.

Answer: (c) Pakistan emits less than 1% of global GHGs but suffers extreme warming and floods. — Pakistan contributes less than 1% of global GHG emissions but is among the top 10 most vulnerable countries.

In the textbook, the options of MCQ 38 are labelled (a), (c), (c), (d) and those of MCQ 43 are labelled (a), (b), (c), (b). The options are lettered here in their printed order as (a)–(d).

studysapiens.com

SECTION 2: SHORT QUESTIONS

1. What are the primary reservoirs of the chemicals of biogeochemical cycles?

A **primary reservoir** ("sink" or "pool") is a massive component of the biosphere where the element of a biogeochemical cycle is stored for long periods of time. On this basis the cycles are classified as:

- **Gaseous cycles:** the reservoir is the air or the oceans (via evaporation). They include the nitrogen, oxygen, carbon and water cycles.
- **Sedimentary cycles:** the reservoir is the Earth's crust. They include the phosphorus, sulfur and calcium cycles.

2. Write notes on de-nitrification and ammonification.

Denitrification is the exit of nitrogen to the atmosphere. In waterlogged soils, denitrifying bacteria (e.g., *Pseudomonas* and *Thiobacillus*) convert nitrates back into nitrogen gas (N_2), which is released into the air. It keeps the cycle balanced.

Ammonification is the recycling step. When plants and animals die, or when animals release nitrogenous waste, decomposers (fungi and bacteria) break down the complex proteins and convert the organic nitrogen back into ammonia.

3. Briefly describe the role of bacteria in the nitrogen cycle. Enlist their names and roles.

Bacteria carry out almost every step of the nitrogen cycle: they fix atmospheric nitrogen, convert ammonia into nitrates that plants can absorb, decompose dead matter, and return nitrogen to the air.

Bacteria	Step	Role
Cyanobacteria, *Azotobacter*, *Clostridium* (free-living)	Nitrogen fixation	Convert N_2 to ammonia
Rhizobium (symbiotic, in root nodules of legumes)	Nitrogen fixation	Convert N_2 to ammonia
Nitrosomonas	Nitrification	Convert ammonia to nitrite
Nitrobacter	Nitrification	Convert nitrite to nitrate
Decomposer bacteria (with fungi)	Ammonification	Convert organic nitrogen to ammonia
Pseudomonas, *Thiobacillus*	Denitrification	Convert nitrates to N_2

4. Write a brief note on the flow of energy in trophic levels.

The flow of energy is unidirectional. Producers (trophic level 1) convert about 1 to 2% of the solar energy falling on them into the bond-energy of organic matter. From each trophic level only about 10% of the energy is transferred to the next; the rest is lost. For example, of 10,000 kcal fixed by producers, primary consumers get 1,000 kcal, secondary consumers about 100 kcal and tertiary consumers about 10 kcal.

5. Write brief notes on population density and population growth.

Population density is the number of individuals per unit area or volume, e.g., the number of mulberry trees per square kilometre in Muzaffarabad, or the number of *E. coli* per millilitre in a test tube. At high density, as a population approaches its carrying capacity, competition for resources becomes severe; the birth rate decreases and the risk of mortality increases. High densities can also cause high rates of pollution.

Population growth is the increase in the number of individuals over a specific time period. There are two growth models:

- **Exponential growth model:** the population grows without limits at its maximal rate. It does not add a constant number each generation; it increases by the same ratio per unit time. Not all populations show it.
- **Logistic growth model:** the growing population eventually reaches a limit due to shortage of resources (space, light, water, nutrients). It stabilizes at a certain size, the **carrying capacity** of the place. The growth curve is sigmoid (flattened S).

6. Investigate the effects of human population growth on the environment.

The rapidly growing human population is perhaps the greatest challenge to the future of the biosphere. Its impacts are:

- Habitat destruction and changes in land use
- Depletion of natural resources (freshwater, fossil fuels etc.)
- Poverty and inflation
- Increased waste generation and elevated pollution
- Climate change
- Deforestation and loss of biodiversity
- Increased chances of new epidemics and pandemics, starvation and malnutrition

7. How does the increased greenhouses gases in atmosphere result in global warming?

Greenhouse gases allow sunlight to pass through but trap the heat radiated from the Earth's surface. Burning fossil fuels and deforestation have greatly increased these gases, particularly CO_2 (from about 280 ppm before industrialization to 428 ppm in 2026). More gases trap more heat, so less heat escapes into space. This intensified (enhanced) greenhouse effect raises the Earth's average surface temperature, which is global warming (about 1.1°C above the pre-industrial era).

8. What are the long-term effects of global warming?

- Melting of glaciers; once they vanish, the Indus River could run dry.

- Rise in sea level and coastal erosion; seawater intrusion turns fertile land into salty marshes (Thatta, Badin).
- Food insecurity: lower yields of wheat and rice; desertification.
- Ecosystem disruption: species change habitat or migrate, become threatened or extinct.
- More frequent heatwaves, droughts, wildfires and intense floods, with soil erosion and contamination of water resources.

9. Describe the unique adaptations of flora in the Mangrove ecosystem.

The mangrove flora (95% *Avicennia marina*, Timmar) lives at the interface of land and sea. Its adaptations are:

- **Salt tolerance:** the trees grow in brackish water, a mix of fresh river water and salty seawater.
- **Breathing roots:** the soil is anaerobic (oxygen-poor) mudflat, so the trees have "breathing roots" that take in air.

10. Distinguish between the Abiotic profiles of a Temperate Deciduous Forest and a Tropical Thorn Forest.

Abiotic factor	Temperate Deciduous Forest	Tropical Thorn Forest
Region	Moist mountain regions (northern areas)	Indus Plains
Climate	Moderate, with distinct seasons and a cold winter	Arid to semi-arid
Temperature	4°C to 30°C	Summer temperatures can reach 50°C
Annual rainfall	750 mm to 1500 mm	Scanty: 200 mm to 750 mm
Soil	Rich, greyish-brown, with high organic matter (humus)	Not described in the textbook

11. Enlist the key fauna and flora of temperate deciduous forests, tropical rain forest, and mangrove forest.

Ecosystem	Key flora	Key fauna
Temperate deciduous forest	*Taxus baccata* (Yew), *Pinus wallichiana* (Blue Pine), *Cedrus deodara* (Himalayan Cedar), *Berberis lyceum* (Sumbal)	Rhesus monkey, black bear, leopard cat, wolves, various birds
Tropical thorn forest	*Prosopis cineraria* (Jhand), *Salvadora oleoides* (Peelu), *Capparis aphylla* (Karir), *Vachellia nilotica* (Kikar)	Camels, houbara bustard, chinkara, golden jackal, Balochistan gerbil, porcupines, rattlesnakes
Mangrove forest	*Avicennia marina* (Timmar), 95% of the forest	Crocodiles, shrimps, mudskippers, crabs, lobsters, turtles, snakes, dolphins; flamingo, crane, pelican

The question prints "tropical rain forest", but the chapter describes no rain forest. The Tropical Thorn Forest (section 25.6.2) is evidently meant and is given here.

12. Differentiate between:

• Biome and biosphere

Biome	Biosphere
A large, distinct geographical region with a specific climate, soil, altitude and adapted plants and animals.	Every part of the Earth where life exists; the global sum of all biomes.
There are many biomes.	There is only one biosphere; the highest and most inclusive level.
Examples: coniferous forest biome (Murree, Swat), desert biome (Thal, Thar).	Extends from the deepest sea trenches to the highest peaks and several kilometres into the atmosphere.

• Biogeochemcial cycle and energy flow

Biogeochemical cycle	Energy flow
Movement of chemical elements (matter) between biotic and abiotic components.	Movement of energy from the sun through the trophic levels.
Cyclic: the elements are used again and again.	Unidirectional: energy is not recycled.
Matter is not lost.	Energy is lost at each trophic level; only about 10% passes on.
Examples: water cycle, nitrogen cycle.	Example: producers → herbivores → carnivores.

• Population growth and population density

Population growth	Population density
Increase in the number of individuals over a specific time period.	Number of individuals per unit area or volume.
Described by the exponential and logistic growth models.	Example: mulberry trees per square kilometre; *E. coli* per millilitre.

• Convex population and concave population

Convex (Type I) population	Concave (Type III) population
Individuals survive to an old age and then die rapidly.	Very high juvenile mortality; adults have a much lower mortality rate.
Environmental factors have very little impact on mortality.	Most individuals die young.
Examples: humans and mammals.	Examples: fishes and many invertebrates.

• Nitrification and nitrogen-fixation

Nitrification	Nitrogen fixation
Conversion of ammonia to nitrite and then to nitrate.	Conversion of atmospheric N_2 into inorganic compounds like ammonia.
By *Nitrosomonas* and *Nitrobacter*.	By cyanobacteria, *Azotobacter*, *Clostridium*, *Rhizobium*; also by lightning and industry.
Makes nitrogen easy for plants to absorb.	Brings nitrogen from the air into the soil.

• Nitrification and denitrification

Nitrification	Denitrification
Ammonia → nitrite → nitrate.	Nitrate → nitrogen gas (N_2).
By *Nitrosomonas* and *Nitrobacter*.	By *Pseudomonas* and *Thiobacillus*, in waterlogged soils.
Makes nitrogen available to plants.	Returns nitrogen to the atmosphere.

SECTION 3: LONG QUESTIONS

31. What is a biogeochemical cycle? Describe water cycle in detail.

Biogeochemical cycles are the cyclic movements of chemical elements, such as carbon, nitrogen, oxygen and phosphorus, between the living (biotic) and non-living (abiotic) components of the ecosystem.

A **primary reservoir** (“sink” or “pool”) is a massive component of the biosphere where the element of a biogeochemical cycle is stored for long periods of time. On this basis the cycles are classified as:

- **Gaseous cycles:** the reservoir is the air or the oceans (via evaporation). They include the nitrogen, oxygen, carbon and water cycles.
- **Sedimentary cycles:** the reservoir is the Earth’s crust. They include the phosphorus, sulfur and calcium cycles.

The water cycle

The **water cycle** (hydrological cycle) is the continuous movement of water between the abiotic components (air, soil and sea) and the biotic components.

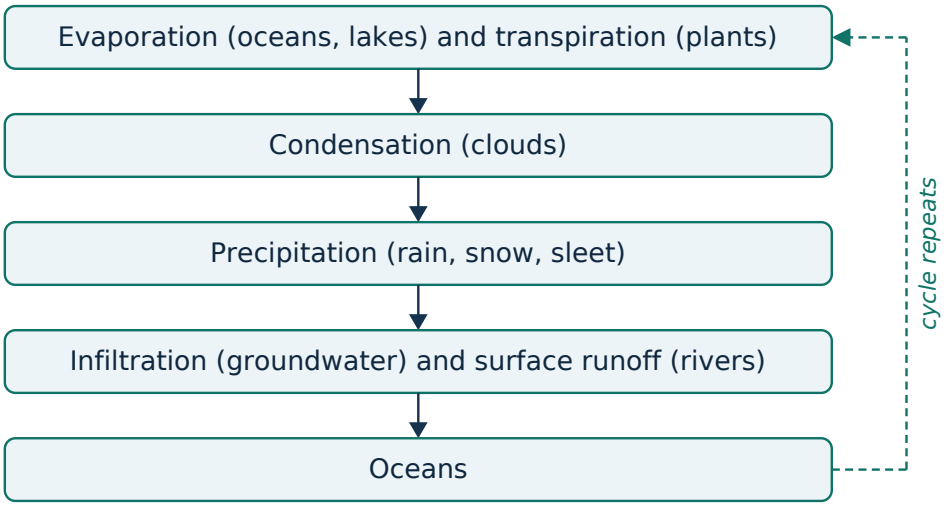
Abiotic loop

1. **Evaporation:** The sun powers the entire cycle. Solar radiation heats the surface water of oceans and lakes, and water molecules evaporate as vapours. About 86% of global evaporation occurs from the oceans (for Pakistan, the Arabian Sea).
2. **Condensation:** Rising vapours lose heat. When the air is cold enough, they turn back into liquid droplets or ice crystals, which cling to dust particles to form clouds.
3. **Precipitation:** When the droplets become too heavy, they fall due to gravity as rain, snow or sleet (rain mingled with snow or hail).
4. **Infiltration and runoff:** In *infiltration*, water seeps into the soil (percolation) and moves down to the water table, where it is stored as groundwater. *Surface runoff* is the unconfined flow of water over the land when it can no longer infiltrate; it flows into streams, rivers and finally the oceans.

Biotic loop

1. **Entry of water:** Terrestrial plants absorb water from the soil through root hairs and use it for photosynthesis and for support (turgor pressure). Animals take in water by drinking and by eating food.
2. **Exit of water:** *Transpiration* through the stomata of leaves is the major exit point. *Respiration* in plants and animals produces “metabolic water”, exhaled as vapours. In *excretion and egestion*, animals release water in urine and faeces. In *perspiration*, water leaves through skin pores and evaporates. *Decomposition* of dead bodies also returns water to the soil or atmosphere.

studysapiens.com



The water cycle

32. Describe nitrogen cycle in detail.

The **nitrogen cycle** is the process by which nitrogen circulates between the biotic and abiotic components of the ecosystem. Atmospheric nitrogen (N_2) is transformed into compounds that living organisms can use, and is later returned to the atmosphere. It has five steps:

1. Nitrogen fixation (entry into the soil)

The atmosphere is 78% nitrogen gas, whose atoms are joined by a strong triple bond that plants cannot break. Nitrogen fixation is the conversion of N_2 into inorganic compounds like ammonia (NH_3). It occurs in three ways:

- **Biological fixation** (more than 90%): by soil microorganisms. *Free-living* (non-symbiotic): cyanobacteria (blue-green algae), *Azotobacter*, *Clostridium*. *Mutualistic* (symbiotic): *Rhizobium* in the root nodules of legumes such as chickpea (chana) and mung bean (moong). They convert N_2 to ammonia.
- **Atmospheric fixation** (5–10%): lightning during rains and storms gives the energy to combine nitrogen with oxygen; N_2 is converted into nitrogen oxides and then into nitrates.
- **Industrial fixation:** the synthesis of nitrogen-containing fertilizers; atmospheric nitrogen and hydrogen are combined to produce ammonia.

2. Nitrification (conversion to plant food)

Ammonia is toxic or difficult for plants to use in large amounts. Soil bacteria convert it: *Nitrosomonas* convert ammonia to nitrite (NO_2^-), and *Nitrobacter* convert nitrite to nitrate (NO_3^-). Nitrates are easily absorbed by plants.

3. Assimilation (entry into the biotic world)

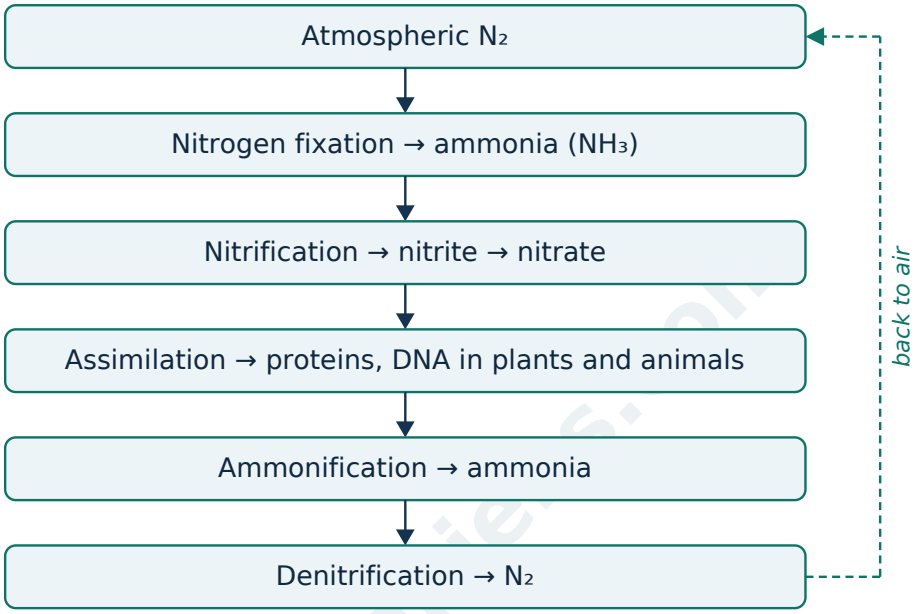
Plants and microbes absorb nitrates from the soil and use them to make proteins, DNA and chlorophyll. Animals assimilate nitrogen by eating plants and other animals.

4. Ammonification (the recycling step)

When plants and animals die, or when animals release nitrogenous waste, decomposers (fungi and bacteria) break down the complex proteins and convert the organic nitrogen back into ammonia.

5. Denitrification (exit to the atmosphere)

In waterlogged soils, denitrifying bacteria (e.g., *Pseudomonas* and *Thiobacillus*) convert nitrates back into nitrogen gas (N_2), which is released into the atmosphere. This keeps the cycle balanced.



The nitrogen cycle

33. Explain the flow of energy in successive trophic levels.

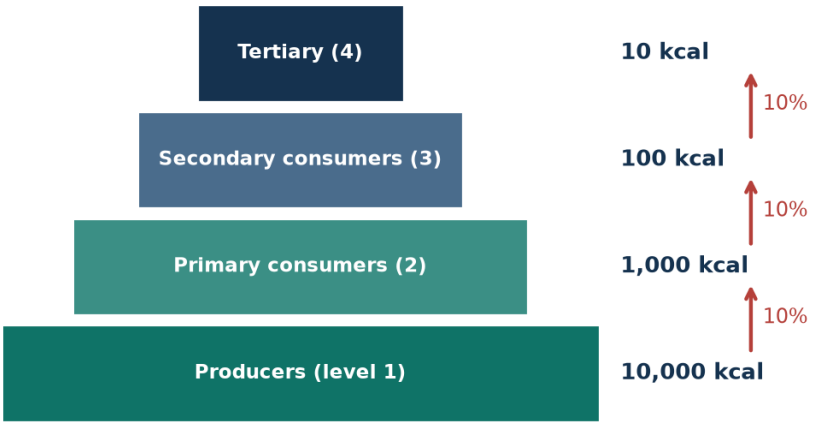
In an ecosystem the organisms are arranged in different feeding groups, known as **trophic levels**.

Trophic level	Organisms	Examples
1	Primary producers	Green plants, algae, photosynthetic bacteria
2	Primary consumers (herbivores); feed directly on producers	Insects, rabbit
3	Secondary consumers (carnivores) and parasites of animals; feed on herbivores	Frog, snake
4	Tertiary consumers (animals that eat carnivores)	Wolf, hawk
Separate level	Decomposers and detritivores; feed on all trophic levels (detritus and faecal matter)	Crabs, vultures, jackals

Loss of energy between trophic levels

Unlike the cycling of matter, the flow of energy is **unidirectional**. There are generally three or four trophic levels in an ecosystem.

- About **1 to 2%** of the solar energy that falls on the producers is converted to the bond-energy of organic matter.
- Only about **10%** of the energy of one trophic level is transferred to the next; energy is lost at each level.
- So very little usable energy remains after it has passed through the bodies of organisms at different trophic levels.
- **Example:** If producers fix 10,000 kcal, the primary consumers get only 1,000 kcal, the secondary consumers about 100 kcal, and the tertiary consumers about 10 kcal.



Loss of energy between trophic levels (textbook example)

Short food chains (fewer trophic levels) are more energy efficient than long ones, because energy is lost as heat during transfer at each trophic level.

34. Explain the problems related to the rapid growth of human populations and the effects of that growth on ecosystem.

Rapid growth of human populations

Starting in the early 1700s, changes in technology gave humans more control over their food supply, superior weapons against predators, and remedies against many diseases. Humans also became less vulnerable to climatic uncertainties. These changes allowed humans to **expand the carrying capacity** of their habitats, so the population grew rapidly.

The world population reached 8 billion in 2022; about 16% lives in developed countries and 84% in developing countries. Pakistan's population was 241.49 million in the 2023 census, growing at 2.55% annually.

Effects on the ecosystem

The rapidly growing human population is perhaps the greatest challenge to the future of the biosphere. Its impacts are:

- Habitat destruction and changes in land use

- Depletion of natural resources (freshwater, fossil fuels etc.)
- Poverty and inflation
- Increased waste generation and elevated pollution
- Climate change
- Deforestation and loss of biodiversity
- Increased chances of new epidemics and pandemics, starvation and malnutrition

35. Describe the causes and impacts of the increasing concentration of greenhouse gases in the atmosphere.

The **greenhouse effect** is the natural process in which gases present in the Earth's atmosphere trap the heat radiated from the Earth's surface. These gases are called **greenhouse gases (GHGs)**. They allow sunlight to pass through but prevent heat from escaping directly back into space, so they keep the Earth warm enough to sustain life.

Causes

There has been a marked increase in the concentration of GHGs, particularly CO_2, mainly due to the use of fossil fuels for running machinery, driving vehicles, heating homes and generating electricity. The enhanced, unnatural warming of the atmosphere caused by excessive GHGs is the major negative impact of human activities on the environment.

Greenhouse gas	Sources	Share of global warming
Carbon dioxide (CO_2)	Burning fossil fuels (coal, oil, gas); deforestation	About 80%
Methane (CH_4)	Production of coal and gas; livestock (enteric fermentation in cattle); agricultural practices	About 11%
Nitrous oxide (N_2O)	Agricultural and industrial activities; combustion of fossil fuels	About 6%
Fluorinated gases (chlorofluorocarbons)	Synthetic; industrial processes (old refrigerators and air conditioners)	About 3%

Impacts

1. **Global warming:** the long-term increase in the Earth's average surface temperature due to human activities. According to the IPCC Report (2023), global surface temperature has risen by roughly 1.1°C compared with the pre-industrial era (1850–1900).
2. **Heatwaves:** cities like Jacobabad and Sibi are among the hottest places on Earth. Above 50°C the human body cannot adjust its temperature, leading to fatal heatstroke; livestock suffer heat stress. Shorter, warmer winters let pests and mosquitoes survive longer. Heat worsens drought and increases the risk of wildfires.

3. **Floods:** a warmer atmosphere holds more moisture, giving erratic and intense rainfall and "turbo-charged" monsoons. Floods damage forests, homes and agriculture, cause soil erosion, nutrient depletion and contamination of water, and spread water-borne diseases.
4. **Glacial melting:** glaciers (like Baltoro and Batura) melt rapidly, causing Glacial Lake Outburst Floods. In the long run, once the glaciers vanish, the Indus River could run dry.
5. **Sea level rise and coastal erosion:** polar ice melts and seawater expands with heat. In Thatta and Badin, seawater intrusion is turning fertile land into salty marshes.
6. **Food insecurity:** high temperature at the flowering stage of wheat and rice reduces grain yield; intense heat evaporates soil moisture and turns fertile land into desert (desertification).
7. **Ecosystem disruption:** rapid shifts in temperature and rainfall threaten biodiversity; species are forced to change habitat or migrate, and may become threatened or extinct.

36. Provide a comprehensive account of the Tropical Thorn Forest of Pakistan. Give its location, key flora and fauna, and its current ecological state.

Feature	Description
Nature	The most widespread terrestrial ecosystem of the plains of Pakistan; xerophytic (drought-resistant) plants with thorns to deter herbivores and reduce water loss
Location	The Indus Plains: much of Punjab (Bahawalpur, Khanewal, Mianwali, Dera Ghazi Khan), Sindh (southern regions bordering the Indus) and Balochistan (western regions)
Climate	Arid to semi-arid; summer temperatures can reach 50°C
Water	Scanty rainfall (200 mm to 750 mm annually)
Key flora	*Prosopis cineraria* (Jhand/Kandi), *Salvadora oleoides* (Peelu/Wan), *Capparis aphylla* (Karir), *Vachellia nilotica* (Kikar)
Key fauna	Camels, houbara bustard, chinkara, golden jackal, Balochistan gerbil, porcupines, rattlesnakes
Current status	**Heavily degraded.** Most of these "natural jungles" have been cleared for human settlements and the canal-irrigated agriculture of the Punjab

37. Write a detailed note on the Temperate Deciduous Forest in Pakistan.

Feature	Description
Nature	Found in moist mountain regions with distinct seasons; a cold winter causes the trees to drop their leaves (deciduous)
Location	Northern areas: Shogran, Neelam Valley, Murree and parts of Kaghan
Soil	Rich, greyish-brown soil with high organic matter (humus)
Climate	Moderate temperatures (4°C to 30°C); annual rainfall 750 mm to 1500 mm
Key flora	*Taxus baccata* (Yew), *Pinus wallichiana* (Blue Pine), *Cedrus deodara* (Himalayan Cedar); shrubs like *Berberis lyceum* (Sumbal)
Key fauna	Rhesus monkey (*Macaca mulatta*), black bear, leopard cat, wolves and various bird species
Current status	**Degraded.** Many large predators have been wiped out; pressure from the lumber industry and agricultural expansion

INQUISITIVE QUESTIONS

18. Justify the fact that humans are responsible for climate change.

- **Rise in CO_2:** before industrialization the CO_2 concentration was about 280 ppm; it is 428 ppm in 2026 (WMO). The rise began with industrialization.
- **Human sources of GHGs:** CO_2 from burning coal, oil and gas (machinery, vehicles, heating, electricity) and deforestation; methane from coal and gas production, livestock and agriculture; nitrous oxide from agriculture and industry; fluorinated gases are entirely synthetic.
- **Matching rise in temperature:** global surface temperature has risen by about 1.1°C compared with 1850–1900 (IPCC, 2023).
- **Consumption:** the wealthiest 20% of the world's population produces 53% of the CO_2 emissions.
- **Growing population:** more people cause more land-use change, deforestation and use of fossil fuels.

Since the extra greenhouse gases come from human activities, and the warming follows them, humans are responsible for the present climate change.

19. Investigate the effects of human population growth on ecosystem.

The rapidly growing human population is perhaps the greatest challenge to the future of the biosphere. Its impacts are:

- Habitat destruction and changes in land use
- Depletion of natural resources (freshwater, fossil fuels etc.)
- Poverty and inflation
- Increased waste generation and elevated pollution
- Climate change

- Deforestation and loss of biodiversity
- Increased chances of new epidemics and pandemics, starvation and malnutrition

How these affect the ecosystem: Clearing land for settlements and agriculture destroys habitats (e.g., the tropical thorn forest cleared for canal-irrigated agriculture; temperate forests under pressure from the lumber industry). Loss of habitat reduces biodiversity and may push populations below their minimal viable size. Burning more fossil fuels adds greenhouse gases and causes climate change, and more waste pollutes soil, water and air (e.g., pollution from Karachi threatening the mangroves).

20. Justify why science education has become necessary for everyone to understand the steps man has to take to save the biosphere.

- The threats to the biosphere (greenhouse gases, global warming, deforestation, loss of biodiversity, pollution) are scientific problems; they cannot be understood without basic science.
- The causes lie in everyday human activities (fuel, electricity, food, waste), so every person, not only scientists, has to change behaviour.
- An educated public can understand data such as CO_2 levels and temperature rise, and can tell facts from rumours.
- People who understand cycles of matter, energy flow and carrying capacity see why resources are limited and why population growth must be managed.
- Science education prepares citizens to support and follow sound policies: saving water and energy, planting trees, protecting wildlife, restoring ecosystems such as mangroves.

21. Investigate the careers related to the study of environmental resources.

Career	Work
Ecologist	Studies the relationships of organisms with one another and with the environment
Environmental scientist / consultant	Monitors pollution and assesses the environmental impact of projects
Wildlife biologist / conservationist	Protects endangered species and manages national parks
Forester (forest officer)	Manages and restores forests
Climatologist / meteorologist	Studies climate, weather and climate change
Hydrologist / water resource manager	Studies and manages rivers, groundwater and irrigation
Marine and fisheries biologist	Studies coastal ecosystems such as mangroves, and fish stocks
Soil scientist / agronomist	Works on soil fertility and sustainable agriculture

Career	Work
Environmental engineer	Designs waste-treatment, clean-water and renewable-energy systems
Demographer	Studies the characteristics of populations

Questions 18–21 need answers beyond the textbook; the figures quoted are from the chapter, while the careers list and parts of Q.20 are general knowledge.

22. Activity: Search internet and collect the pictures of the key fauna and flora of different ecosystems in Pakistan.

This is a practical activity to be done by the student. Search for and collect pictures of the following (names from the textbook), and paste them in your notebook under the four headings:

Ecosystem	Flora to search	Fauna to search
Temperate deciduous forest	Yew, Blue Pine, Himalayan Cedar (Deodar), Sumbal	Rhesus monkey, black bear, leopard cat, wolf
Tropical thorn forest	Jhand, Peelu (Wan), Karir, Kikar	Camel, houbara bustard, chinkara, golden jackal, Balochistan gerbil, porcupine
Alpine and sub-alpine meadows	Juniper, Birch, willow, alpine grasses and wildflowers	Snow leopard, Himalayan brown bear, Himalayan ibex, musk deer, marmot, snowcock, monal, golden eagle
Mangrove forest	*Avicennia marina* (Timmar)	Crocodile, shrimp, mudskipper, crab, lobster, turtle, dolphin, flamingo, crane, pelican